Fiscal Year:	FY 2013	**Task Last Updated:**	FY 08/03/2012
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Project Title:	Effects of Space Radiation on the Function of Human Brain Capillaries and the Blood-brain-barrier: Implications for Neurodegenerative Disorders		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	10032-3702	**Congressional District:**	13
Comments:			
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No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Ponnaiya, Brian (Columbia University) Smilenov, Lubomir (Columbia University)		
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Task Description:	This proposal is aimed at determining the effects of space radiation on the function of human brain blood vessels (capillaries) and the Blood-Brain-Barrier (BBB). The brain vascular system is crucial to healthy functioning of the brain and its dysfunction is not only a primary event in a range of neurodegenerative diseases but also an important influencing factor in many others. Furthermore, vessel pathologies involve common susceptibility factors such as oxidative stress and inflammation so any radiation effects on capillaries can be influenced by other preexisting conditions. Here, we propose to investigate the effects of space radiation on angiogenesis and BBB function using human brain blood vessel cells in 2-dimensional and 3-dimensional tissue models. Since the brain vascular system is essentially an integrated multi-cell system we also propose to utilize other human brain cell types in tissue models to create the most realistic human vessel model possible. We have focused on the main aspects of vessel function that are relevant to the research emphases of acute and late risks to the brain. That is, functioning of the BBB, blood vessel development and the compound effects of preexisting conditions such as inflammation. A correctly functioning blood system supplies the brain and other tissues with oxygen and nutrients while preventing deleterious blood-borne molecules from entering the brain and causing possible damage to brain cells. Tight junctions between the blood vessel cells are formed by proteins that make a tight seal and prevent leakiness. We will determine the effect of space radiation on blood vessels of the brain and its possible impact on neurodegeneration by testing for vessel formation (angiogenesis) vessel leakiness, and the influence of other factors such as inflammation molecules after exposure to both galactic cosmic radiation and radiation from the sun.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Gamma radiation studies Studies to optimize assay conditions using gamma radiation revealed a previously unknown effect of gamma photons on human blood vessels. It is the first demonstration of a transient and rapid effect of gamma radiation on human endothelial barriers that involves cell detachment and loss of the cell adhesion molecule - PECAM-1. This effect occurred at the fractionated radiotherapy dose of 2 Gy and therefore has the potential to be of clinically relevant importance considering the role of PECAM-1 in many of the cellular processes involved in pathologies seen at longer times after radiation therapy. As such it could be a target for treatments that alleviate these pathologies. A second potential benefit is that of drug delivery to tumors. A temporary opening of the endothelial barrier by the dual effects of low temperatures and gamma radiation could be a means by which impermeable chemotherapy drugs could be delivered directly to tumors. Tissue models The development of 3-Dimensional human tissue models from normal human cells and stem cells has great potential in many fields of medical research. Tissue models can more accurately depict human tissue since the cells can be arranged spatially as they would be in vivo and can interact with each other as they would in the human body. A neurovascular unit can be used for basic research on many aspects of the human brain. These include regeneration, synaptic function, and degeneration. Because the tissue model is derived from individual cells, each cell type can be altered genetically before it is incorporated into the model.
Task Progress:	Endothelial barrier studies Gamma radiation While waiting for beam time we tested the assays using Gamma radiation. This was intended for several reasons. It allowed us to allow us to determine how the endpoints relate to each other with regard to sensitivities. We were also able to streamline the assays before going to NSRL. In addition, it gave us a counterpoint to compare the effects of ion particles with. Finally, our investigations of the early effects of photons revealed the first demonstration of a transient and rapid effect of gamma radiation on human endothelial barriers that involves cell detachment and loss of the cell adhesion protein - PECAM-1 (or CD31). Such an effect has the potential to be clinically relevant to longer-term effects of radiotherapy. This work has been submitted to the International Journal of Radiation Oncology, Biology, Physics. Ion particles Determination of the threshold levels for various space particle radiations has begun in the summer run of 2012 and will continue probably through 2013 before the effects on TEER, endothelial barrier permeability, and junction proteins in human 2D and 3D tissue models have been established. We have shown that the effects small ion particles are not the same as gamma photons and that they cause a long-term degradation of endothelial barrier function in cells from umbilical cord and from brain capillaries. The effects of a larger particle – Silicon has also been investigated and found to have a different effect again on TEER. Stained samples are still being processed from this series of experiments. Angiogenesis studies We have previously shown that low LET protons and high LET Fe ions inhibit the formation of brain capillaries by different mechanisms. In an ongoing study to determine the relationship between LET and the inhibition of angiogenesis we have examined the effects of Silicon ions that have an intermediate LET. Samples are being processed at the time of writing but preliminary observations indicate that Silicon ions behave like Fe ions. Other Studies We have initiated an in vivo study in collaboration with Dr. Richard Britten to investigate the brain vasculature of rats exposed to space radiations. Initially we will look for abnormalities in fixed brains from Dr. Britten’s experiments.
Bibliography Type:	Description: (Last Updated: 03/04/2024)

Articles in Peer-reviewed Journals	Grabham P, Bigelow A, Geard C. "DNA damage foci formation and decline in two-dimensional monolayers and in three-dimensional human vessel models: Differential effects according to radiation quality." International Journal of Radiation Biology. 2012 Jun;88(6):493-500. Epub 2012 Apr 30. PMID: 22449005 , Jun-2012